Elevated monoamine oxidase a levels in the brain: an explanation for the monoamine imbalance of major depression.

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CONTEXT: The monoamine theory of depression proposes that monoamine levels are lowered, but there is no explanation for how monoamine loss occurs. Monoamine oxidase A (MAO-A) is an enzyme that metabolizes monoamines, such as serotonin, norepinephrine, and dopamine. OBJECTIVE: To determine whether MAO-A levels in the brain are elevated during untreated depression. SETTING: Tertiary care psychiatric hospital. PATIENTS: Seventeen healthy and 17 depressed individuals with major depressive disorder that met entry criteria were recruited from the care of general practitioners and psychiatrists. All study participants were otherwise healthy and nonsmoking. Depressed individuals had been medication free for at least 5 months. MAIN OUTCOME MEASURE: Harmine labeled with carbon 11, a radioligand selective for MAO-A and positron emission tomography, was used to measure MAO-A DVS (specific distribution volume), an index of MAO-A density, in different brain regions (prefrontal cortex, anterior cingulate cortex, posterior cingulate cortex, caudate, putamen, thalamus, anterior temporal cortex, midbrain, hippocampus, and parahippocampus). RESULTS: The MAO-A DVS was highly significantly elevated in every brain region assessed (t test; P=.001 to 3x10(-7)). The MAO-A DVS was elevated on average by 34% (2 SDs) throughout the brain during major depression. CONCLUSIONS: The sizable magnitude of this finding and the absence of other compelling explanations for monoamine loss during major depressive episodes led to the conclusion that elevated MAO-A density is the primary monoamine-lowering process during major depression.